

Abstract Linear Algebra

Prof. Lanier

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Pick up a handout. Try those warm-up problems.

Next up:

- Midterm revisions due tonight.
- Quiz 9 today (eigenvalues and eigenvectors.)
- HW₇ assigned today. Last homework, half-sized.
- Problem session on Friday, 3-5pm in Ryerson 177.
- Requizzing session on Sunday, 3-4pm in Cobb 115.
- HW₆ (written and WeBWorK) due on Monday.
- HW₆ debrief/prebrief on Monday, 3:30-4:30 in Eckhart 207A;
No other office hours on Monday.

Let V be a real or complex vector space.

A norm takes one vector as its input and gives a real number length as output.

An inner product takes two vectors as its input and gives a field element as output.

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Every inner product gives a norm, by plugging in the same vector for both inputs.

The standard inner product is the dot product; it gives the Euclidean (L^2) norm.

Properties of general inner products on vector space V

1. (Conjugate) symmetry: $(\mathbf{x}, \mathbf{y}) = \overline{(\mathbf{y}, \mathbf{x})}$; note, that for a real space, this property is just symmetry, $(\mathbf{x}, \mathbf{y}) = (\mathbf{y}, \mathbf{x})$;
2. Linearity: $(\alpha\mathbf{x} + \beta\mathbf{y}, \mathbf{z}) = \alpha(\mathbf{x}, \mathbf{z}) + \beta(\mathbf{y}, \mathbf{z})$ for all vector $\mathbf{x}, \mathbf{y}, \mathbf{z}$ and all scalars α, β ;
3. Non-negativity: $(\mathbf{x}, \mathbf{x}) \geq 0 \ \forall \mathbf{x}$;
4. Non-degeneracy: $(\mathbf{x}, \mathbf{x}) = 0$ if and only if $\mathbf{x} = \mathbf{0}$.

Examples:

The dot product

A scaling of the dot product by a non-zero scalar

The dot product modified by a (conjugate) symmetric square matrix with positive eigenvalues.

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1. All inner products are given by this last construction.
2. For any given inner product on $\mathbb{R}^n$ or $\mathbb{C}^n$, its matrix is just the identity matrix for the right choice of basis.

Orthogonality: vectors, bases, subspaces

The dot product tells us the (scaled) projection of u onto v .

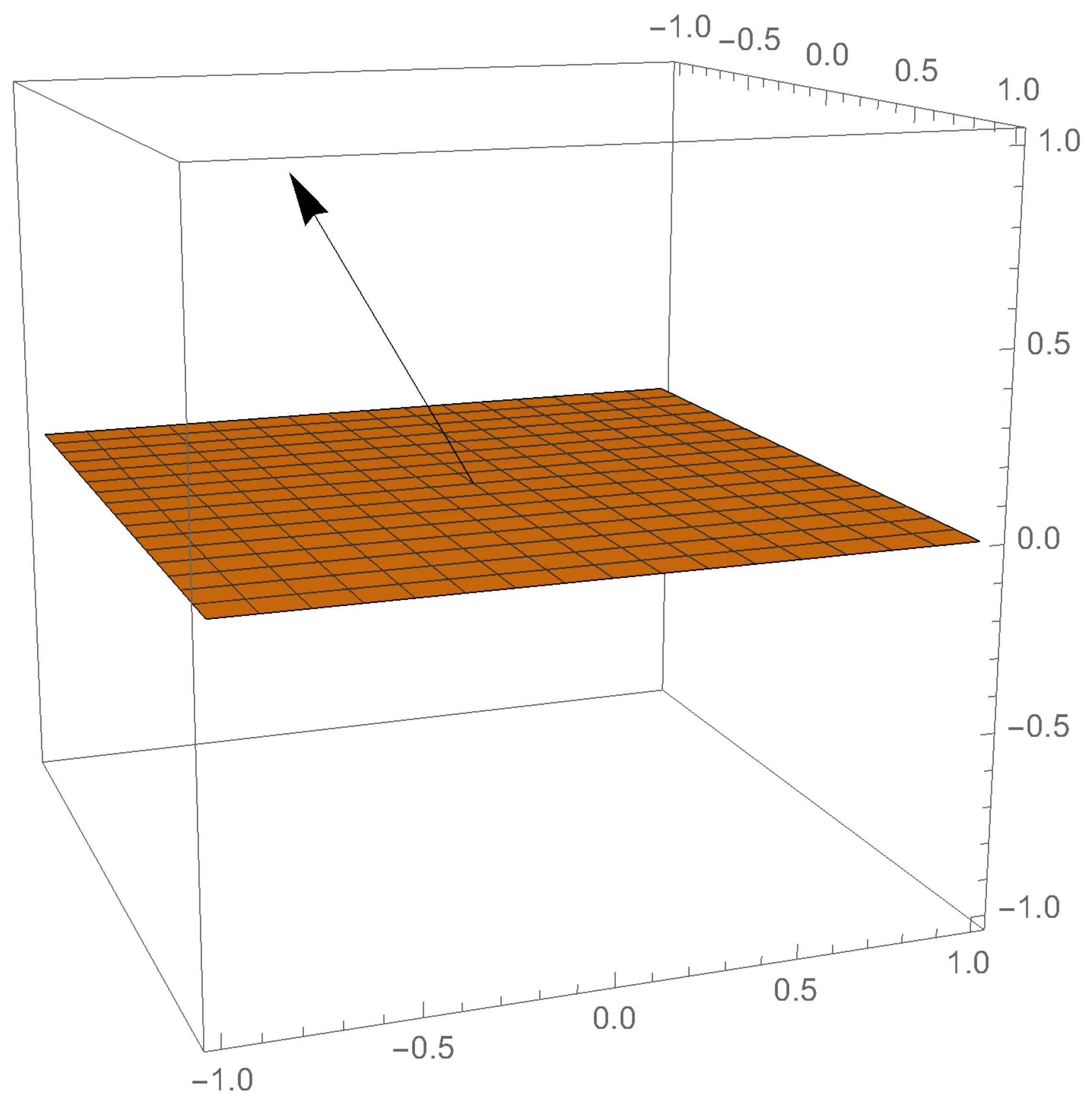
$$u \cdot v = \|u\| \|v\| \cos \theta$$

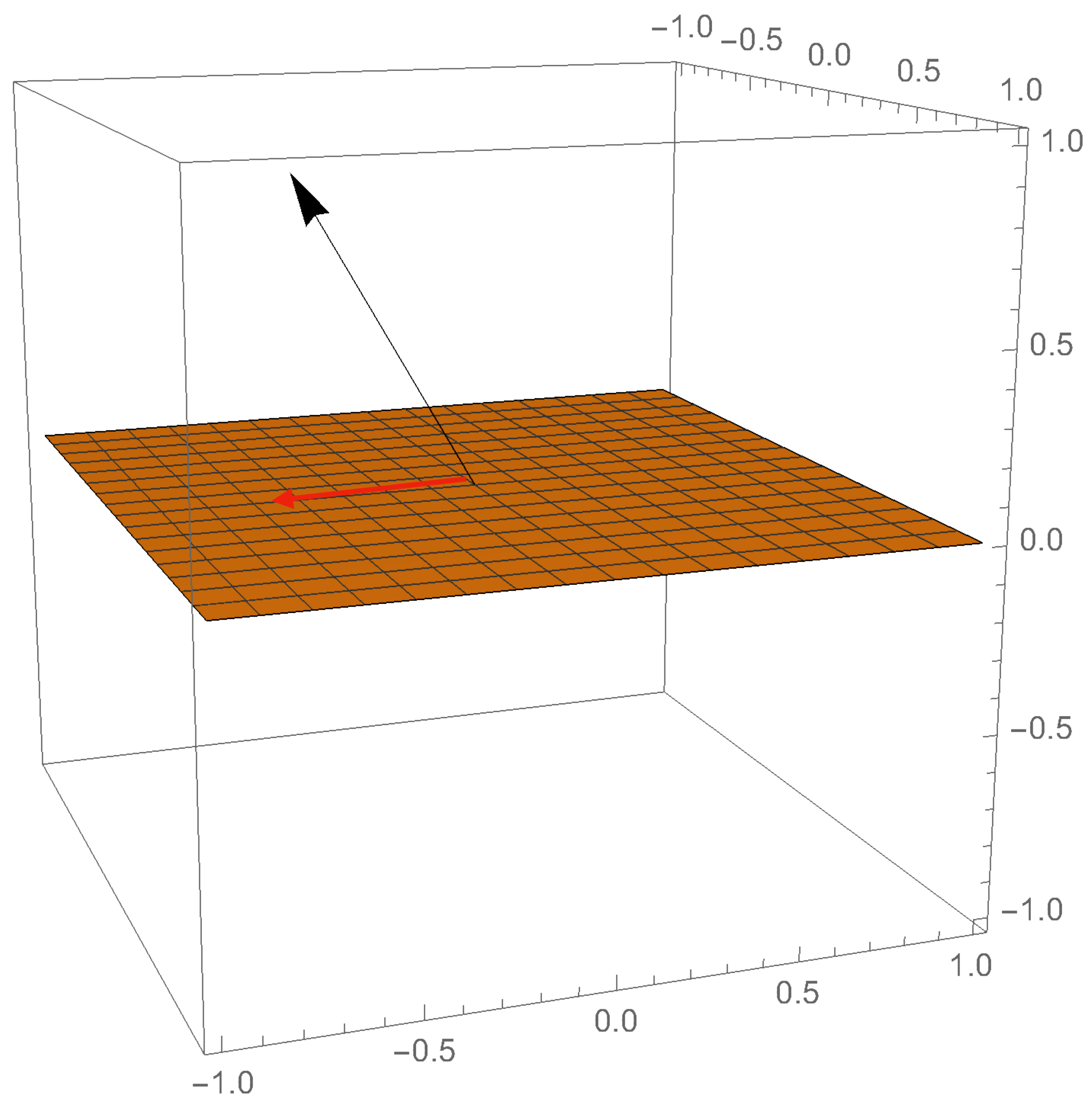
If the dot product of two vectors is 0, then the vectors are orthogonal.

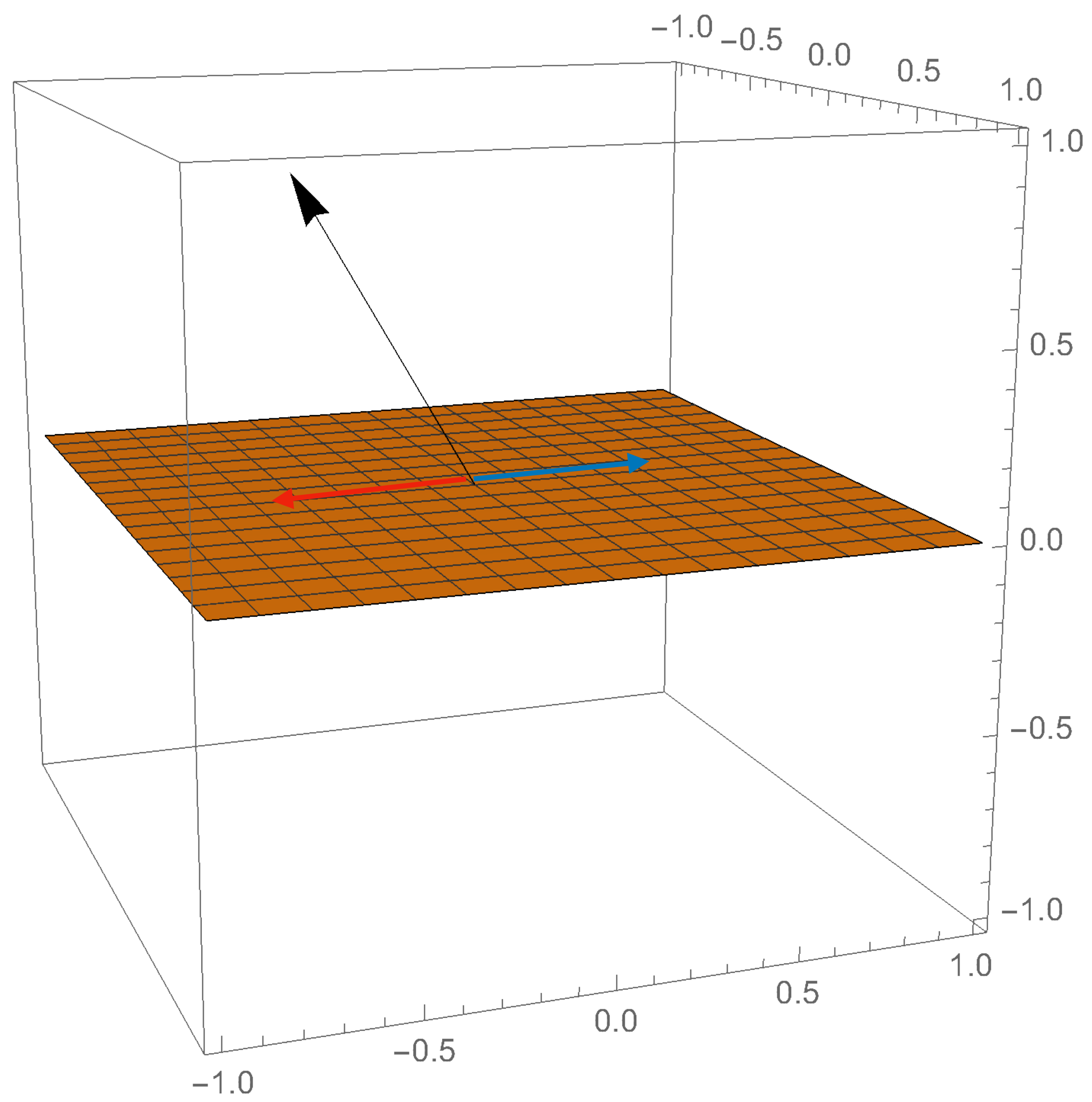
(pairwise)

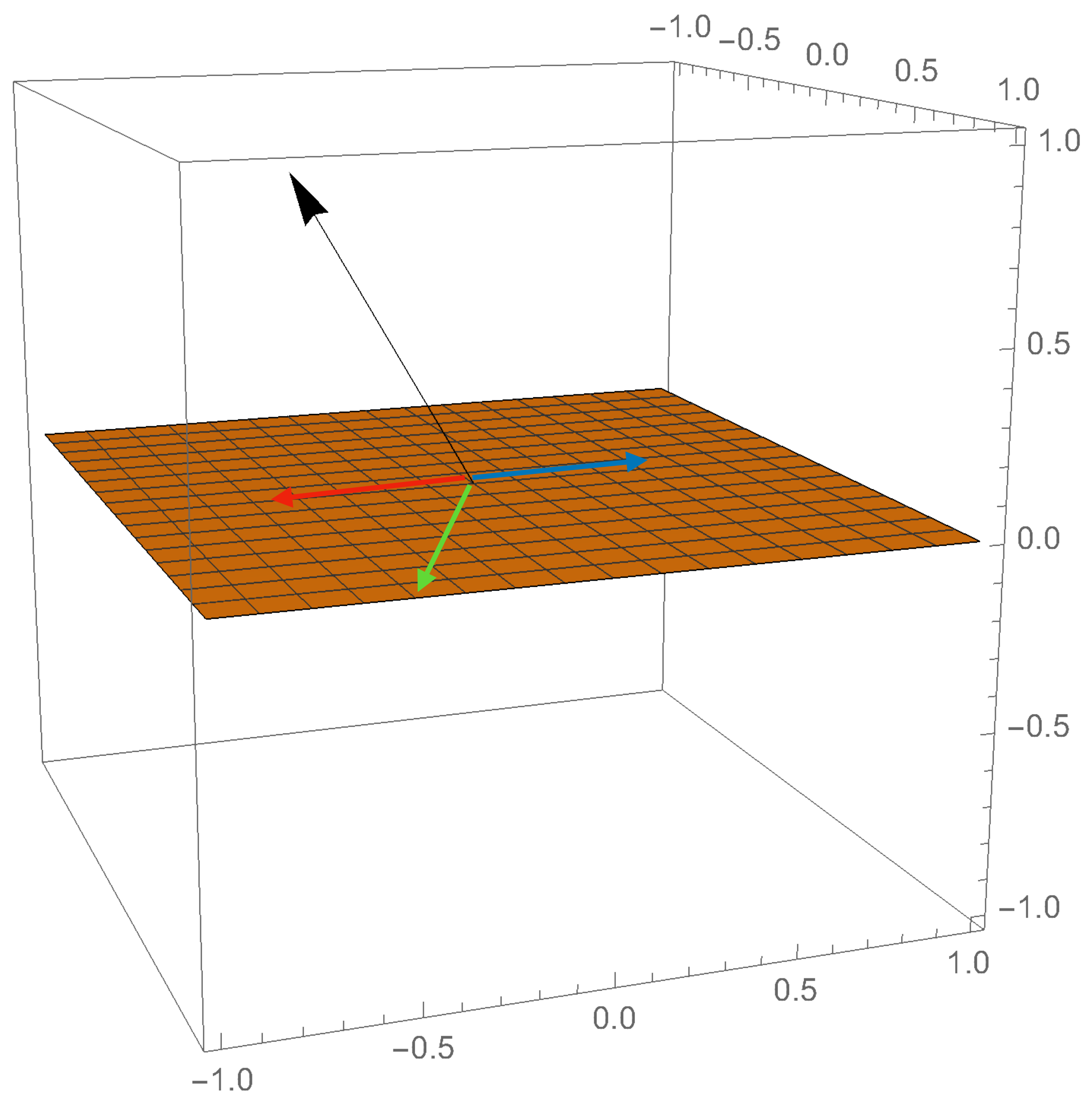
If all of the vectors in a basis are orthogonal, then the basis is orthogonal.

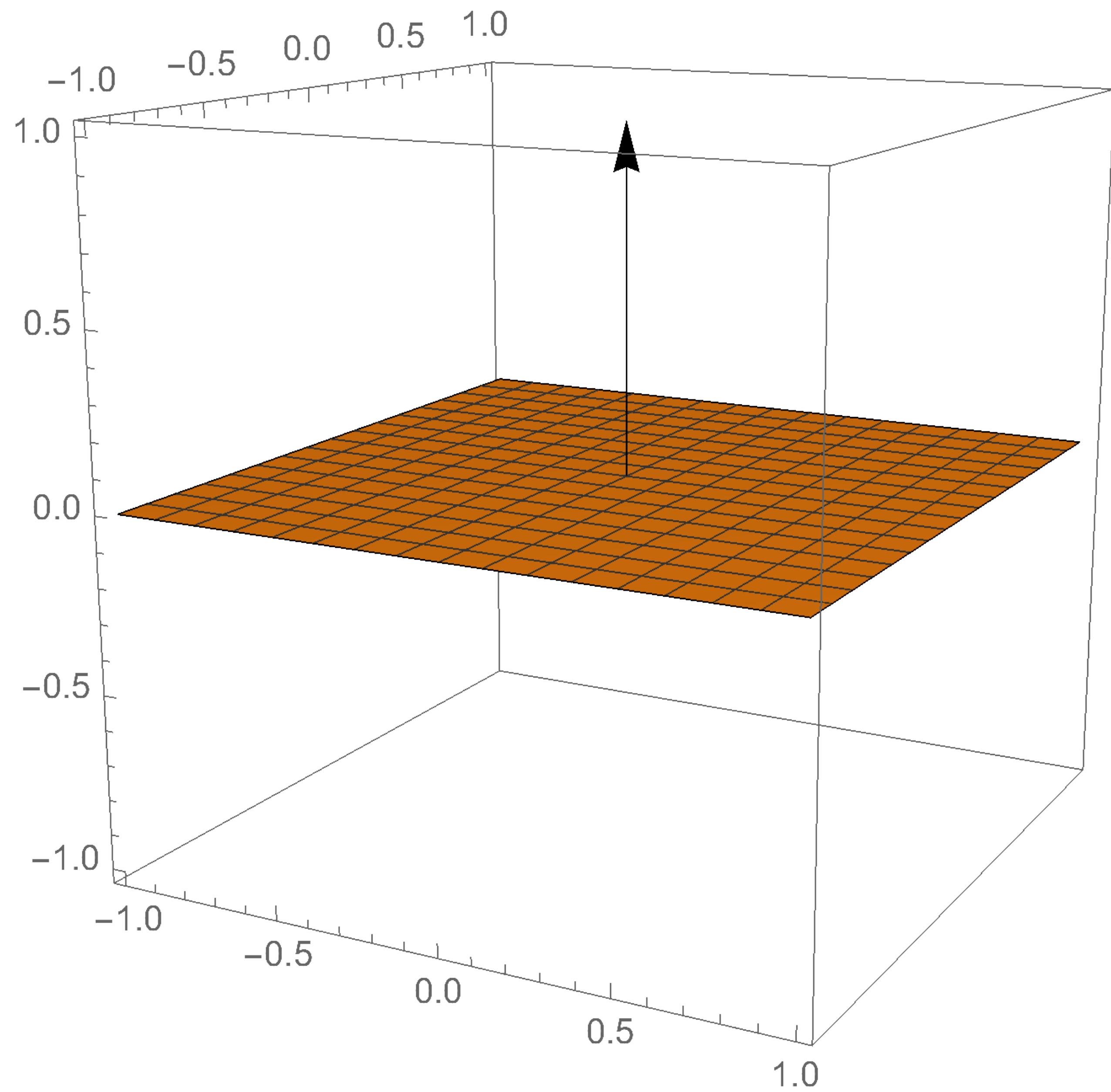
If two subspaces each have a basis where the vectors in one are orthogonal to all the vectors in the other, the subspaces are orthogonal.

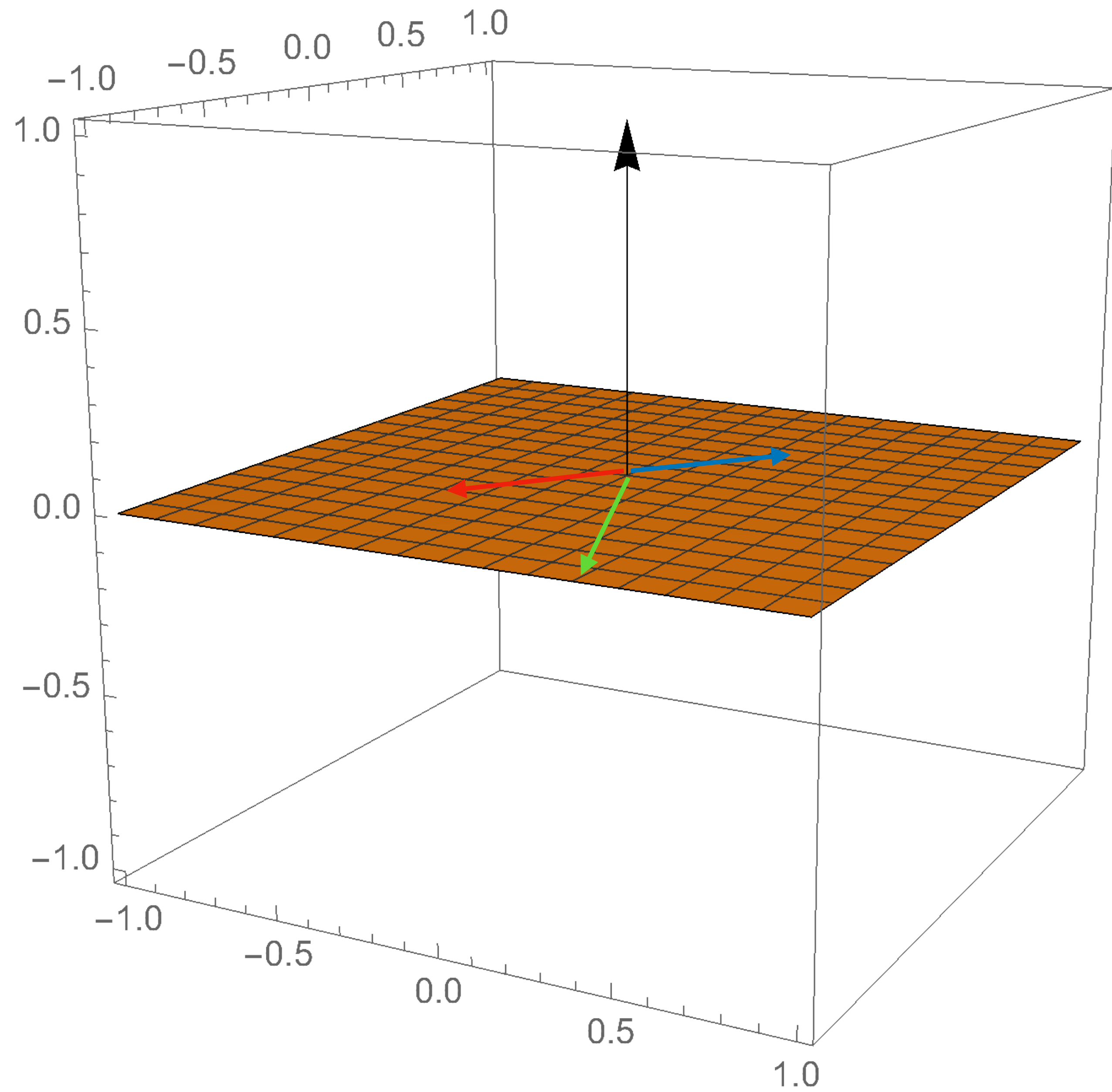












Orthonormal basis: basis vectors are all orthogonal and of unit length.

For metric purposes, an orthonormal basis is just as good as the standard basis.

A matrix with columns an orthonormal basis is invertible and preserves distance/norm/standard inner product.

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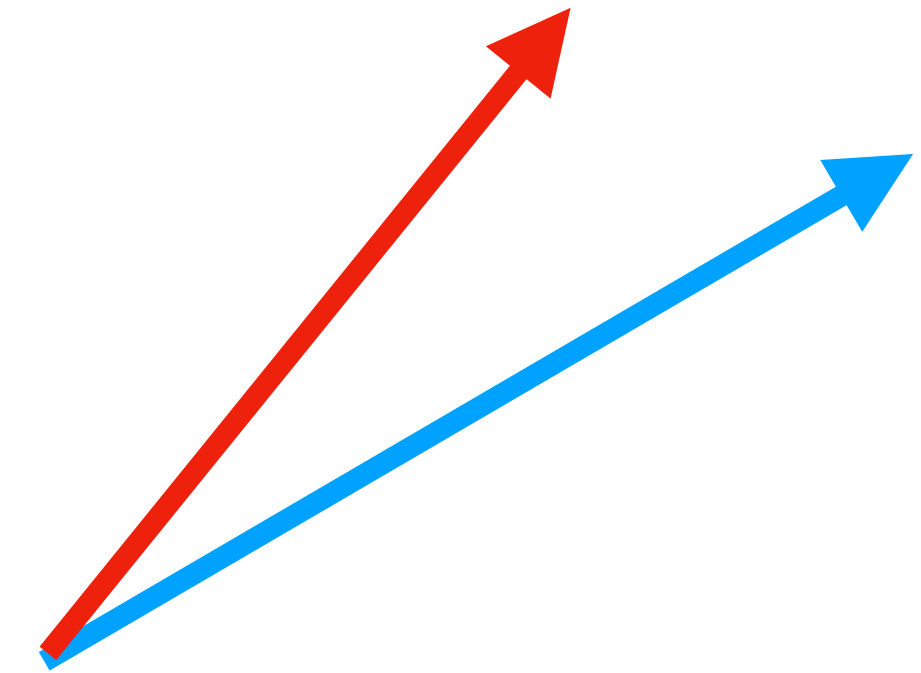
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The Gram-Schmidt process

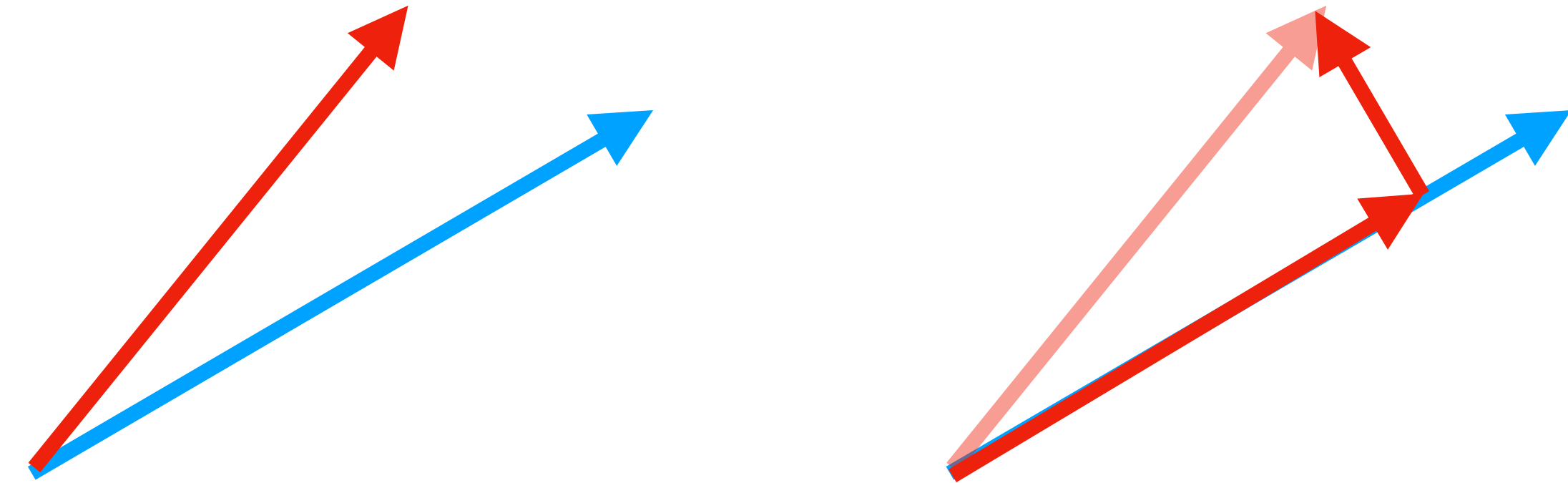
Gram–Schmidt process:

Modify vectors in turn to make them orthogonal to the previous basis vectors,
and then normalize them to give them unit length.



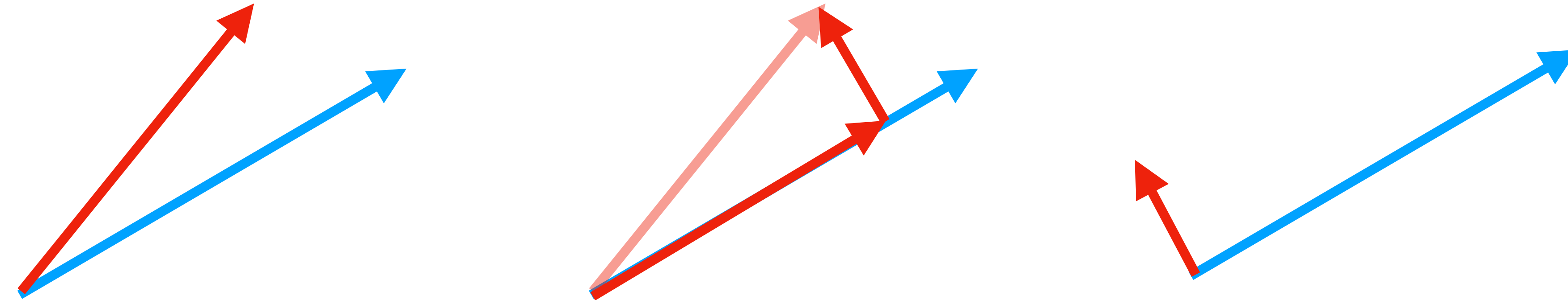
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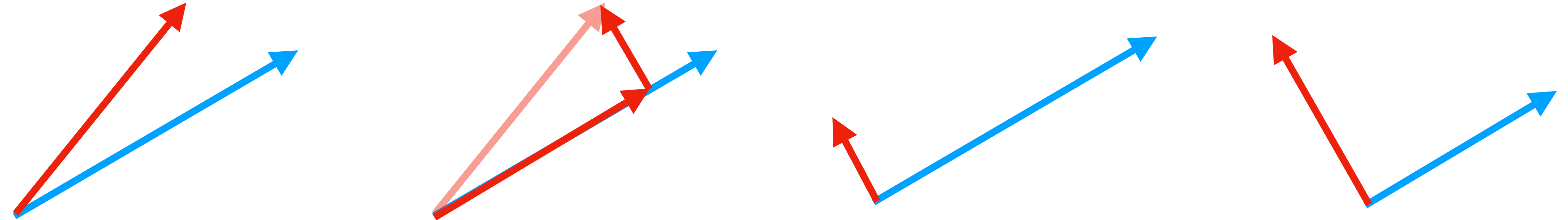
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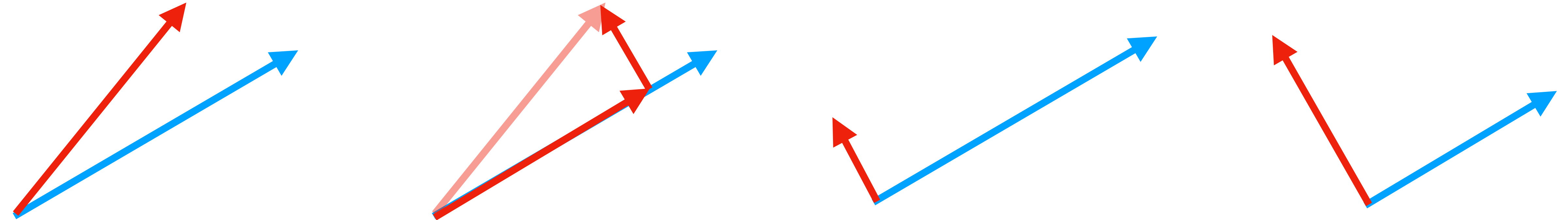
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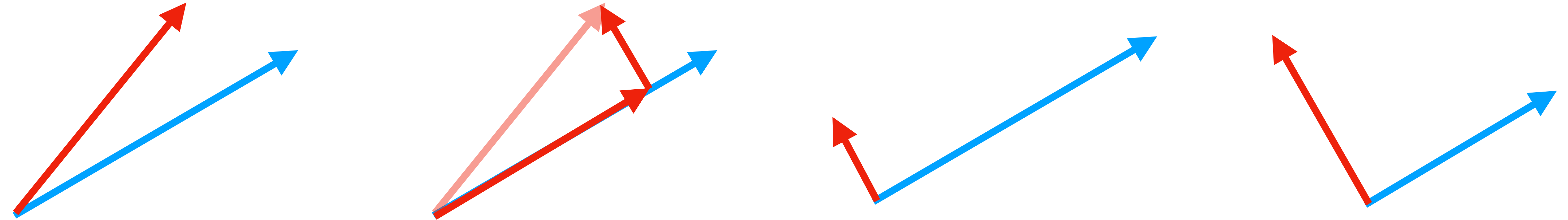


$$\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \quad \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

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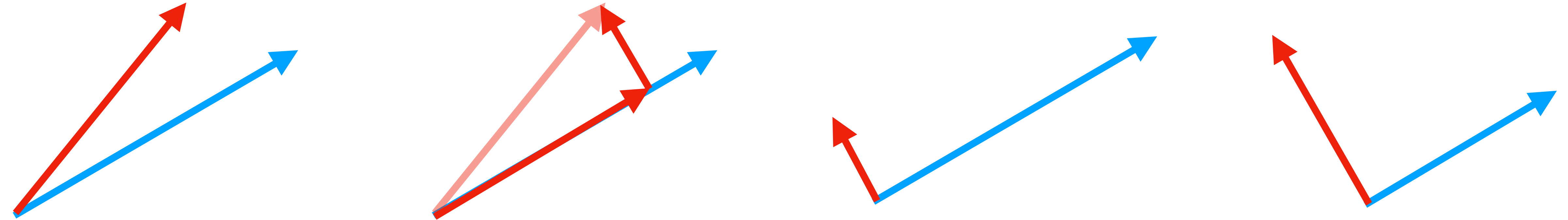
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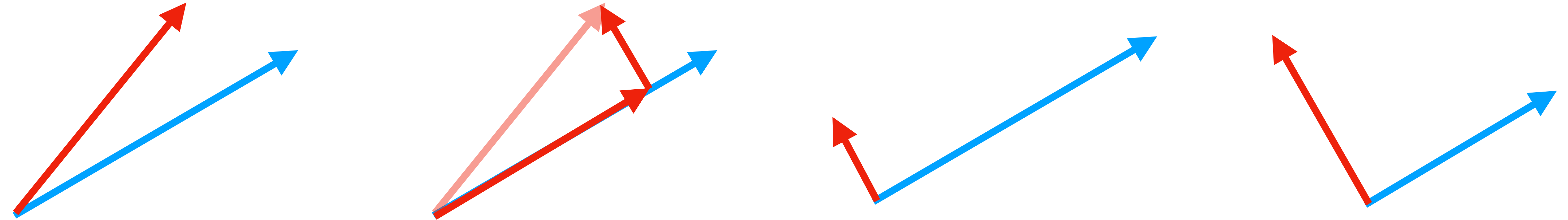
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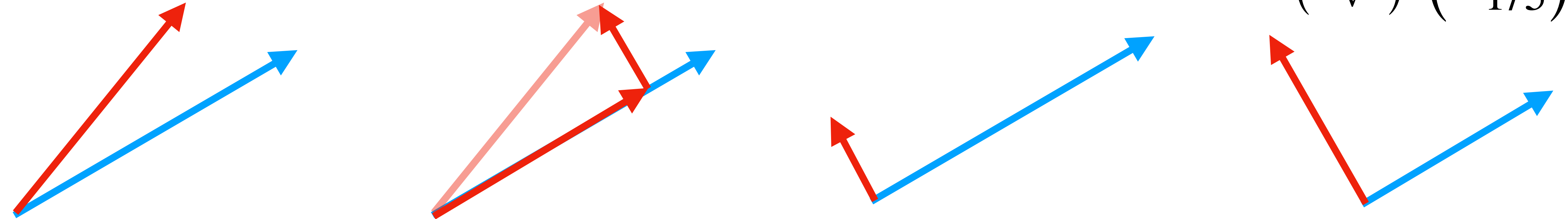
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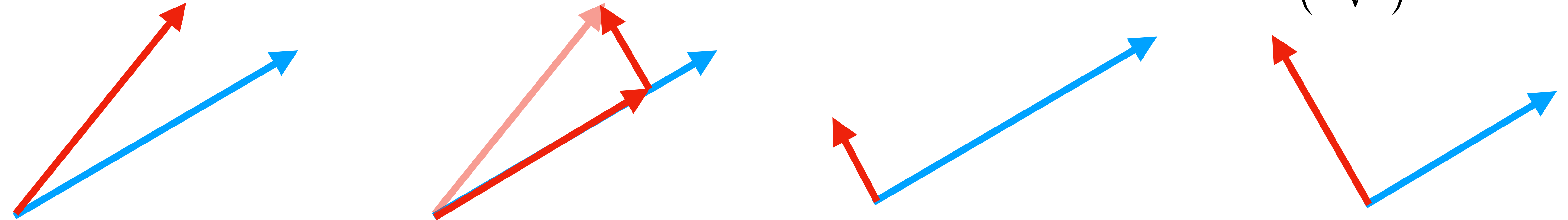
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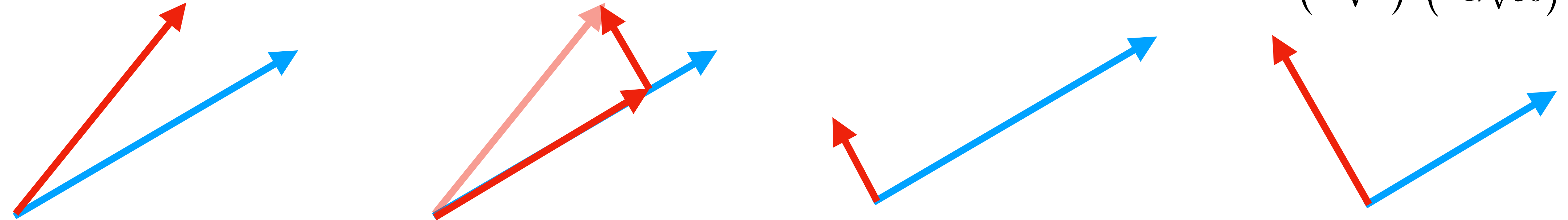
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